Technical review

Item #3 · 20-date-after-first-hop.eml · Run ID: 20261113T232800Z-08b8d1

Path	`R:\evidence\mail\20-date-after-first-hop.eml`
Size (bytes)	1464
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.eml
Detected type (signature)	Email message (RFC 5322)
Status	Needs attention
Case	RV-SAMPLE-001
Evidence ID	E0017
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`4bbb81d83fb442fbf181cc581dd73f8bbb3c7ce41842b629bc44161d7e6fdbfa`
md5	`827870dedf658a5968a9a8beb4900baf`

Findings

[Medium] DKIM signature does not verify (DKIM_SIGNATURE_INVALID)

What it shows	The body still matches the DKIM signature of example.org (selector s2026), but the signature over the signed header fields does not verify with the supplied public key (key source: key_file; verifier: Signature verification failed). Either signed header fields changed after signing, or the supplied key is not the one that signed (for example a rotated key).
What it does not prove	It does not tell which of the two happened, nor who changed a header or why; with a rotated or wrong key a genuine message gives the same result.

[Informational] Receiving server's checks recorded (AUTH_RESULTS_RECORDED)

What it shows	1 Authentication-Results header(s); the top one, written by mx.acme.example, states: dkim=pass, spf=pass.
What it does not prove	These are claims written into the message by a server; anyone handling the file could have added or edited them, and they were not re-checked.

[Medium] Date header later than the first server (DATE_AFTER_FIRST_HOP)

What it shows	The Date header (2026-11-10T08:47:00Z) is 2820 s later than the time the first server received the message (2026-11-10T08:00:00Z).
What it does not prove	The sender's clock may be wrong; it does not show that the date was changed after sending.

Email analysis

Message-ID	20261110090000.1@example.org
From	Bruno Proveedor <bruno@example.org>
Return-Path	bruno@example.org
Subject	Acta de la reunión de noviembre
Date header	Tue, 10 Nov 2026 09:47:00 +0100 (2026-11-10T08:47:00Z)

DKIM mode / key file	offline · R:\inputs\dkim-keys.txt · 4 key(s)

Received hops (first server = 1)

Hop 1	From host: [192.0.2.10] · Received by: smtp.example.org · Date stamped: 2026-11-10T08:00:00Z
Hop 2	From host: smtp.example.org · Received by: mx.acme.example · Date stamped: 2026-11-10T08:00:04Z

DKIM signatures (offline verification)

#1 d=example.org s=s2026	Result: signature_invalid · Algorithm: rsa-sha256 · Key source: key_file (line 3) · Signature verification failed
ARC chain	none (0)

Authentication-Results (as stated by the servers)

mx.acme.example	dkim=pass, spf=pass

Files written by the analysis

headers.json (headers)	md5 9d96f70c94dcf8cb66535ef61468cf3b · sha256 6f2b686543c2503520ccebe566431def70c3acad92fbbaf6a71e09ea8289e1ab
dkim.json (dkim)	md5 aa8ee8943df2e6f503b988ed875667e5 · sha256 9b943b8e76557ad9093ab439498873fcb6536be2bd41cf8a99bb92c4751c749f

Method

Each file was read once to compute its digests, then analysed as email: .eml as an RFC 5322 message, .msg through its MAPI properties, and each message of an mbox as an item of its own (each in a separate worker process with time and memory limits when worker isolation is on, as recorded in the audit log). The header section, Received hops and dates were read and compared; DKIM and ARC signatures were verified offline with the public keys the examiner supplied (no DNS query unless online mode was chosen); Authentication-Results were recorded as stated evidence; attachments were hashed; and BEC heuristics compared the addresses with the organisation's own domains. Every step and every file written is recorded in the hash-chained audit log.

Audit trail

Audit entries	#37 – #43
Log folder	`R:\case\runs\rv-mailauth\20261113T232800Z-08b8d1\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- DKIM results depend on the public keys supplied: a check is only meaningful with the key that was valid when the message was sent. Keys are rotated, so a current key can fail a genuine message and a missing key leaves the signature unchecked.
- A missing DKIM signature is common and proves nothing; a valid signature covers only the signed header fields and the body, not who wrote the message.
- Outlook .msg files do not keep the original message: the body is re-encoded (so DKIM cannot be checked) and the original header section is kept only when transport headers were stored.
- Lookalike-domain, Reply-To, display-name and payment-wording checks are heuristics with false positives and false negatives; the organisational domain comes from the bundled Public Suffix List, whose version is recorded in environment.json and may differ from the list in force when the message was sent.

- Received, Date and Authentication-Results headers are text that servers and clients wrote; unsigned headers can be added or edited by anyone who handled the file.
- In an mbox, body lines starting with "From " are escaped with '>'; un-escaping follows the selected variant (mboxrd by default) and cannot always restore the original line, which can affect DKIM body hashes.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.